Inferring When to Help in Proactive AI Assistance

PhD research proposal | Jianxuan Shao | 22 October 2026

Research motivation and primary contribution

Repeated changes in a visual editor can mean that someone is struggling, exploring alternatives, or deliberately practising. An assistant may identify a useful next step while misunderstanding how much help the person wants. I will investigate how assistants can decide whether a specific suggestion should appear before an explicit request. The main question is whether explicitly representing task relevance separately from willingness to receive a particular form of help improves that decision.

Task relevance means that a feasible suggestion addresses the current artefact or operation. Receptivity means that this person would welcome this content, in this form, now. A relevant suggestion may be unwelcome because the user wants to practise, retain creative control, or continue without interruption. This distinction does not establish actual benefit: a welcome suggestion could still harm learning or produce a poor result. Both willingness and task outcomes therefore need evaluation.

The same pause, revision or undo sequence can accompany difficulty, deliberate practice or exploration, yet imply different willingness to receive help. Treating these cues as a general signal to intervene could interrupt independent work and leave users repeatedly dismissing suggestions or explaining their goals. A unified model could learn the distinction from context and feedback. The question is whether explicit separation helps avoid these errors under limited development data, and whether any predictive advantage translates into better interaction.

The primary test will compare welcome predictions, with calibration and error analysis helping explain any difference. Mixed-initiative and notification research establishes contextual trade-offs [1] [2]; DesignScape also shows tension between useful suggestions and users' control [3].

A concrete task and interaction

In the proposed testbed, participants will create an event poster from supplied text and images. In a production task, they will aim to complete a readable layout; in a practice task, they will learn an unfamiliar alignment operation. The assistant will observe the stated goal, selected objects, their positions, and recent moves, alignment commands and undo events inside the editor. It will use these bounded workspace records rather than infer emotions or private intentions from a camera.

Suppose a participant repeatedly adjusts the spacing of three text boxes. The assistant may prepare a short hint, 'Select the boxes and use Distribute vertically', or an editing proposal with a preview of equal spacing. The user must confirm any edit. The assistant can also wait. Repeated undo will trigger assessment, not automatically trigger a message. The two suggestions can both address the same operation while differing in whether the user wants them. During independent practice, a preview may reveal more of the solution than the user wishes to receive.

In Study 3, a user might decline that preview and say, 'I want to practise this myself.' This will count against similar editing proposals in the current practice task, while leaving the operation relevant. On returning to a production task, the same person may welcome a preview. The system must reconsider the current goal rather than turn one refusal into a permanent preference. This is a proposed interaction to investigate, not an assumed pattern in users' behaviour.

Closest work and the remaining empirical question

ProMemAssist subtracts working-memory displacement and interference costs from assistance importance and relevance, deciding to deliver, defer or suppress a message [4]. It already compares structured timing with direct LLM judgement. Its implemented costs concern cognitive processing; my target is whether a specified intervention is welcome. A broader utility model can include agency and practice costs, so my value-cost comparator will receive those same goal and preference records.

Sensing What Surveys Miss adjusts assistance thresholds from acceptance and task correctness, including reducing help after rejection despite an incorrect answer [5]. It also collects independent reports of needing help. Expanded robot-assistance theory considers skill and social costs [6]. These precedents establish that inferred need and willingness can differ; the open comparison concerns how well explicit structure predicts welcome for a particular action.

After Talking with 1,000 Personas uses five preference dimensions, category-supervised population training and individual activation steering [7]. Adaptation uses accept/reject/ignore feedback, dimension ratings and optional free-text feedback, with category-specific directions and strengths and repeated storyboard evaluation. I will build on this rich feedback design by comparing representations during active editing with equal access to the original reasons.

RLVF's C3PO constructs in-scope, near-scope and out-of-scope examples to reduce feedback overgeneralisation [8]. EOPA reuses temporal/activity feedback evidence and calibrates initiation thresholds [9]; a recent position paper distinguishes preference likelihood from its reliability [10]. These are foundations for the supporting repeated-use study, rather than additional mechanisms claimed as new.

ACKnowledge makes plans sensitive to what is acceptable in a particular setting [11]. Research on human signals to robots shows why behaviour must be interpreted in context [12], while work with screenwriters documents how people regulate and sometimes withdraw from AI assistance [13]. These studies motivate testing whether an assistant can distinguish a useful next step from a form of help the person welcomes, especially during practice or creative work.

Research questions and prediction

RQ1. When do technically relevant suggestions receive different welcome judgements across delivery forms and work goals? RQ2. Does explicit separation improve welcome prediction over equally informed direct and value-cost models, and does this translate into better interaction? RQ3. Does any advantage persist when both policies can learn from the same kinds of reason feedback across sessions?

I predict the largest advantage where visible difficulty and the preferred degree of assistance diverge. This is a hypothesis about a modelling constraint under limited development data. An equally informed unified model may already learn the distinction, while the constraint may discard useful context. Equivalent forecasts, or gains removed by equal calibration, would weaken the case for separation.

Study 1 Understanding when suggestions are welcome

Approximately 18 to 24 adults will complete counterbalanced poster production and practice tasks. This formative study will refine the task and measurements. A shared generator will prepare hints and previewable edits at sampled pauses, revisions and task boundaries. Participants will state their task goals and preferences for independent work before judging suggestions. Each candidate record will link the proposed help to the preceding editor events, its delivery form and the stated goal.

Two independent raters will assess feasibility and technical relevance. Participants will separately judge whether each specified suggestion would have been welcome. Recording-based review will reduce interruptions; selected immediate reports will check discrepancies. Candidates will include unsurfaced opportunities as well as delivered offers, avoiding evaluation only where a policy chose to act. These are reported preferences, with recall limitations, rather than access to private intent. Participant and task partitions will keep adjacent actions from leaking across development and evaluation.

Study 2 Predicting welcome and testing practical effects

The structured model borrows the supervised concept-bottleneck principle [14]. Task-side observations will pass through an explicit relevance estimate, which a small welcome predictor combines with the task goal, delivery form and permitted preference history. A unified model receives the same observations and relevance labels without this required bottleneck. Both forecast the probability that this person would welcome this delivery now; relevance will neither gate nor multiply that probability.

Offline comparisons will also include a strong context-aware direct model and a desktop value-cost adaptation. All receive the same candidates, evidence, auxiliary labels, backbone and matched capacity, tuning and calibration budgets. Value-cost scores enter probability comparisons only after a development-fitted mapping to the same welcome target; otherwise they are evaluated as decision policies. Development data alone will select and freeze the strongest eligible probability predictor, the strongest practical policy, preprocessing and thresholds before testing.

A new cohort completing the same tasks will supply untouched common-candidate prediction data under frozen models. Predictions will be logged before welcome reports, which remain research-only and unavailable to either policy. The primary endpoint is the participant-averaged paired Brier-score difference: whether predicted welcome probabilities agree better with users' reports. Sampling will target the precision of this difference using formative variability and a prespecified meaningful improvement. Reliability curves and calibration slopes will diagnose calibration separately, because Brier also reflects discrimination. Goal and delivery-form boundaries remain exploratory unless interaction estimates have adequate precision.

This cohort will be randomised between the proposed policy and the frozen strongest practical comparator. Both can wait, offer a peripheral hint, or show an edit preview requiring confirmation. Users can request help at any time; those requests will be logged separately. Shared development targets for suggestion frequency will limit gains from simply remaining silent. Practical success requires fewer unwanted offers with blinded poster-quality ratings within a prespecified acceptable-loss margin. Pilot event rates and repeated-task variability will inform sample-size simulation for these outcomes; inadequate precision will warrant exploratory conclusions.

Users will review whether delivered suggestions were welcome at delivery. Ignoring or declining alone will not define unwanted help, and missing ratings will remain missing, with sensitivity analyses. Actual suggestion rates, computational cost and an unaided practice-transfer item will be secondary outcomes. Better welcome forecasts will therefore be distinguished from benefits to task performance or learning [15] [16] [17].

Study 3 Adapting after users explain a refusal

A supporting study over four sessions in two weeks will compare explicit relevance/receptivity updates with a strong unified reason-conditioned updater. Participants will be randomly assigned to the two updater conditions for all four sessions. Both groups can accept, decline or add a short reason. Both updaters can read the original reason text and consider timing, autonomy, relevance and contextual scope, with matched history, retrieval and computation budgets. Each group uses its own live history. A paired replay on identical held-out histories and reasons will additionally isolate processing differences; masking reasons will estimate the information benefit separately.

The practice-to-production change described above will anchor the analysis of feedback use. A temporary refusal should not automatically suppress relevant help after a changed goal. Repeated unwanted offers, inappropriate transfer of old reasons, task quality and correction effort will show whether the main finding survives repeated use. Feedback scoping is an established problem; this study tests the proposed representation within that problem.

Expected contributions, feasibility and scope

1. Measurement protocol: A documented procedure will pair independent assessments of technical relevance with action-specific welcome reports, alongside separate task-quality and practice-transfer measures. It will specify sampling, annotation and disagreement handling, making the evaluation reusable in comparable tasks.

2. Comparative evidence: Matched comparisons will show whether explicit separation improves predictions and practical outcomes, and examine where it helps or adds little value. A claim of limited value will require sufficient precision to exclude a meaningful improvement; an imprecise result will remain inconclusive.

3. Interaction-design guidance: Repeated-use evidence will inform how assistants should respond to refusal reasons as users' goals change, including when to reduce or reconsider help. Recommendations will state their task boundaries and the uncertainty in interpreting feedback.

My capstone model comparisons and error analysis, together with Game Fellows prototyping and study preparation, support implementation and evaluation. Existing models and bounded editing operations avoid foundation-model pretraining. Year one covers formative work; year two, modelling and live comparison; year three, repeated use; year four, synthesis. Ethics review, voluntary disclosure and participant-controlled retention precede recruitment.

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